12 Fig. 12 shows the circle $(x-1)^2+(y+1)^2=25$, the line $4y=3x-32$ and the tangent to the circle at the point A (5, 2). D is the point of intersection of the line $4y=3x-32$ and the tangent at A.

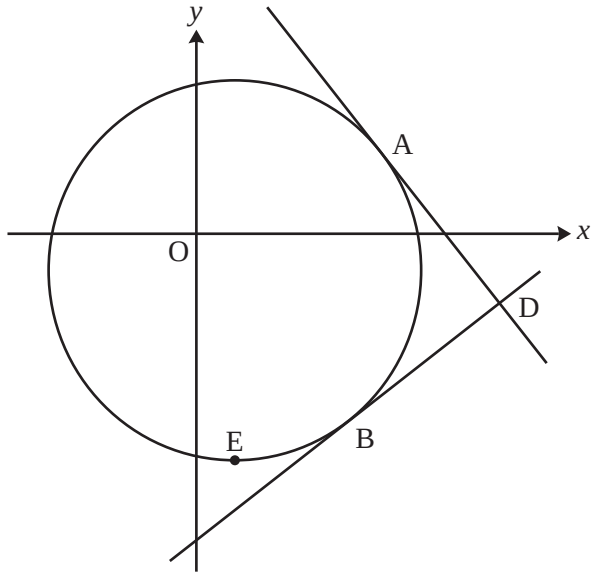

Fig. 12

(i) Write down the coordinates of C, the centre of the circle. **[1]**

(ii) (*A*) Show that the line $4y=3x-32$ is a tangent to the circle. **[4]**

(*B*) Find the coordinates of B, the point where the line $4y=3x-32$ touches the circle. **[1]**

(iii) Prove that ADBC is a square. **[3]**

(iv) The point E is the lowest point on the circle. Find the area of the sector ECB. **[5]**